

Worksheet — Percentage and Its Applications (Class 10 NIOS Mathematics)

Total marks: **30** · Questions: **11**

Instructions: Attempt all 10 questions. Show all working for the subjective questions. Use the formulas precisely; round monetary answers to two decimals. Total time: 50 minutes.

Q1. Express $\frac{1}{8}$ as a percent. [1 mark]

- (A) 8 %
- (B) 10 %
- (C) 12.5 %
- (D) 20 %

Q2. Discount is computed on the marked price. [1 mark]

- (A) True
- (B) False

Q3. If 30 % of $x = 60$, then $x = \underline{\hspace{2cm}}$. [1 mark]

Q4. A 10 % increase followed by a 10 % decrease gives a net change of: [2 marks]

- (A) 0 %
- (B) 1 % decrease
- (C) 1 % increase
- (D) 20 % decrease

Q5. If CP = ₹500 and SP = ₹560, the profit % is: [2 marks]

- (A) 10 %
- (B) 12 %
- (C) 15 %
- (D) 20 %

Q6. A salary of ₹30000 is increased by 12 %. Find the new salary. [2 marks]

Q7. A radio is sold for ₹540 at a gain of 8 %. Find the cost price. [3 marks]

Q8. A shopkeeper marks his goods 25 % above CP and gives a 20 % discount. Find his profit %. [3 marks]

Q9. Find the simple interest on ₹2000 at 8 % p.a. for 3 years. [3 marks]

Q10. Find the compound interest on ₹2000 at 10 % p.a. for 3 years (yearly compounding). Verify that CI > SI for the same data. [6 marks]

Q11. Mira deposits ₹20000 at 8 % p.a. compounded HALF-YEARLY for 1.5 years. Find the maturity amount and the compound interest. [6 marks]

Answer Key

Worksheet — Percentage and Its Applications (Class 10 NIOS Mathematics)

Q1. Express $\frac{1}{8}$ as a percent.

Answer: 12.5 %

$$\frac{1}{8} = 0.125 = 12.5 \%$$

MCQ · EASY · 10-NIOS-MATH-PCT-01

Q2. Discount is computed on the marked price.

Answer: True

Discount reduces MP to SP.

TRUE_FALSE · EASY · 10-NIOS-MATH-PCT-06

Q3. If 30 % of $x = 60$, then $x = \underline{\hspace{2cm}}$.

Answer: 200

$$x = 60/0.30 = 200.$$

FILL_BLANK · EASY · 10-NIOS-MATH-PCT-02

Q4. A 10 % increase followed by a 10 % decrease gives a net change of:

Answer: 1 % decrease

$$1.10 \cdot 0.90 = 0.99 \rightarrow 1 \text{ % net decrease.}$$

MCQ · MEDIUM · 10-NIOS-MATH-PCT-04

Q5. If CP = ₹500 and SP = ₹560, the profit % is:

Answer: 12 %

$$\text{Profit} = 60; 60/500 \times 100 \% = 12 \%$$

MCQ · EASY · 10-NIOS-MATH-PCT-05

Q6. A salary of ₹30000 is increased by 12 %. Find the new salary.

Answer: 30000 · 1.12 = ₹33600.

Use the $(1 + p/100)$ multiplier.

SHORT_ANSWER · EASY · 10-NIOS-MATH-PCT-03

Q7. A radio is sold for ₹540 at a gain of 8 %. Find the cost price.

Answer: CP = 540 / 1.08 = ₹500.

$$\text{CP} = \text{SP} / (1 + \text{gain}\%/100).$$

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-PCT-05

Q8. A shopkeeper marks his goods 25 % above CP and gives a 20 % discount. Find his profit %.

Answer: Let CP = 100. MP = 125. SP = $125 \cdot 0.80 = 100 = \text{CP}$. Profit = 0; profit % = 0.

Markup and discount factors multiply.

SHORT_ANSWER · MEDIUM · 10-NIOS-MATH-PCT-06

Q9. Find the simple interest on ₹2000 at 8 % p.a. for 3 years.

Answer: SI = $2000 \cdot 8 \cdot 3 / 100 = ₹480$.

Plug into SI = $\text{Prt}/100$.

SHORT_ANSWER · EASY · 10-NIOS-MATH-PCT-07

Q10. Find the compound interest on ₹2000 at 10 % p.a. for 3 years (yearly compounding). Verify that CI > SI for the same data.

Answer: A = $2000 \cdot (1.1)^3 = 2000 \cdot 1.331 = ₹2662$. CI = $2662 - 2000 = ₹662$. SI for same data = $2000 \cdot 10 \cdot 3 / 100 = ₹600$. CI – SI = ₹62 > 0; CI exceeds SI by ₹62, the interest-on-interest piece.

CI exceeds SI for $t > 1$ year; the difference is the interest earned by previously credited interest.

LONG_ANSWER · MEDIUM · 10-NIOS-MATH-PCT-08

Q11. Mira deposits ₹20000 at 8 % p.a. compounded HALF-YEARLY for 1.5 years. Find the maturity amount and the compound interest.

Answer: Half-yearly: $r/2 = 4\%$; 1.5 yr → 3 periods. A = $20000 \cdot (1.04)^3 = 20000 \cdot 1.124864 \approx ₹22497.28$. CI = ₹2497.28.

$A = P \cdot (1 + r/(100 \cdot n))^{(n \cdot t)}$ with $n = 2$, $t = 1.5 \rightarrow$ exponent 3.

LONG_ANSWER · HARD · 10-NIOS-MATH-PCT-08